

Advanced Econometrics I

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Week 4 Lecture

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Ordinary Least Squares Estimation

Conditional Expectation Function

Causal Effects

Linear Regression Model

Finite Sample Properties of the OLS Estimator

Gauss Markov Theorem

Weighted and Generalized Least Squares Estimation

We have extensively studied projections of $Y \in L_2$ on the space spanned by $X_1, \dots, X_K \in L_2$

This linear projection has the following minimization problem representation:

$$\begin{aligned}\hat{Y} &:= \operatorname{argmin}_{Z \in \operatorname{sp}(X_1, \dots, X_K)} \|Y - Z\| \\ &= \operatorname{argmin}_{b_1, \dots, b_K} \|Y - (b_1 X_1 + \dots + b_K X_K)\|\end{aligned}$$

Letting $X = (X_1, \dots, X_K)$, why limit ourselves to the subspace $\operatorname{sp}(X)$?

How about this more flexible problem:

$$\operatorname{argmin}_{g(X) \in L_2} \|Y - g(X)\|,$$

where g is any function from $\mathbb{R}^K \rightarrow \mathbb{R}$ with $g(X) \in L_2$

Clearly, the latter minimization problem contains the former

It plays an important role and gets a familiar label:

Definition (Conditional Expectation)

Let $Y, X_1, \dots, X_K \in L_2$ and $X = (X_1, \dots, X_K)$. Let G be the space of functions $g : \mathbb{R}^K \rightarrow \mathbb{R}$ with $g(X) \in L_2$.

Then the **conditional expectation** is defined by

$$\mu(X) := \operatorname{argmin}_{g(X) \in L_2} \|Y - g(X)\|.$$

The set $\{g(X) : g \in G\}$ is a closed subspace of L_2

Think of it as the set of all square-integrable random variables that can be *written as a function of X*

The projection theorem therefore applies, and therefore $\mu(X)$ exists and is unique

The conditional expectation $\mu(X)$ is usually denoted $E(Y|X)$

To be pedantic:

- $E(Y|X)$ is the **conditional expectation** of Y given X
- μ is the **conditional expectation function**

For our purposes, we don't need to worry too much about these nuances

Can regard them as pretty much the same thing

We typically have available random variables Y_i and random vectors X_i with $\dim(X_i) = K \times 1$

Define $\mu(X_i) := E(Y_i|X_i)$

Notice that $\mu(X_i) \in L_2$

We could define $e_i := Y_i - \mu(X_i)$

This is referred to as the CEF error (or simply error term)

This implies the following representation: $Y_i = \mu(X_i) + e_i$

Two useful properties of conditional expectation follow from the projection:

- **Linearity:**

$$E(aY_i + bW_i|X_i) = aE(Y_i|X_i) + bE(W_i|X_i)$$

because projection is a linear operator

- **Conditional determinism or taking out what is known:**

$$E(h(X_i)|X_i) = h(X_i) \text{ for any } h(X_i) \in L_2$$

because $h(X_i)$ already lives in the subspace we are projecting onto

Applying both:

$$\begin{aligned} E(e_i|X_i) &= E(Y_i - \mu(X_i)|X_i) \\ &= E(Y_i|X_i) - E(\mu(X_i)|X_i) \\ &= \mu(X_i) - \mu(X_i) \\ &= 0 \end{aligned}$$

By the law of iterated expectations: $E(e_i) = E(E(e_i|X_i)) = 0$

It follows that $E(e_i) = E(e_i|X_i)$

That is, the conditional mean equals the unconditional mean

This is called **conditional mean independence** or just **mean independence**: information about X_i is irrelevant for the purpose of determining the expected value of e_i

Similar to the case of the linear projection model, the statement

$$Y_i = m(X_i) + e_i, \quad E(e_i|X_i) = 0$$

is not restrictive at all

It tells you that the function $m(X_i)$ must be the CEF $\mu(X_i)$

By definition, the CEF minimizes the distance to Y_i . By how much?

Pick arbitrary $h \in G$, then with $Y_i = \mu(X_i) + e_i$,

$$\begin{aligned}\|Y_i - h(X_i)\|^2 &= E\left((Y_i - h(X_i))^2\right) = E\left(e_i + (\mu(X_i) - h(X_i))\right)^2 \\&= E(e_i^2) + 2 \cdot E(e_i(\mu(X_i) - h(X_i))) + E((\mu(X_i) - h(X_i))^2) \\&= E(e_i^2) + E((\mu(X_i) - h(X_i))^2) \\&= E((Y_i - \mu(X_i))^2) + E((\mu(X_i) - h(X_i))^2) \\&= \|Y_i - \mu(X_i)\|^2 + \|\mu(X_i) - h(X_i)\|^2\end{aligned}$$

notice, the third equality is an application of LIE:

$$E(e_i(\mu(X_i) - h(X_i))) = E((\mu(X_i) - h(X_i)) \cdot E(e_i|X_i)) = 0$$

The first term $\|Y_i - \mu(X_i)\|^2$ is the irreducible prediction error (the best we can do)

The second term $\|\mu(X_i) - h(X_i)\|^2$ is the **excess** prediction error from using h instead of μ

When trying to explain Y_i using X_i

- $\mu(X_i)$ is the best predictor of Y_i
in contrast:
- $X_i'\beta^*$ is the best linear predictor of Y_i

Let's turn to the practical implication next

If an oracle offered us either $X_i'\beta^*$ or $\mu(X_i)$ which one would you prefer to have?

Similarly, if an oracle offered us either a good estimator of $X_i'\beta^*$ or a good estimator of $\mu(X_i)$ which one would you prefer to have?

This raises the question:

What is our overall objective anyway?

Why are we running regressions?

I'm not sure that any group of econometricians (or economists) could agree on a common objective

I'll dip my toe into the waters...

...many econometricians are interested in **causal effects**!

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What is a causal effect?

As good econometricians, we must have some kind of economic thinking that informs us about the relationship between variables

In other words, we have in mind some kind of *structural economic model*

Definition (Structural Model)

Let $X_1, \dots, X_M$ be an exhaustive list of variables that explain Y .

A **structural model** is given by

$$Y = h(X_1, \dots, X_M),$$

where $h(\cdot)$ is a well behaved function.

Given a structural model, we are now in a position to define precisely what we mean by *causal effect*

Definition (Causal Effect)

The **causal effect** of X_i on the outcome variable Y in the structural model $Y = h(X_1, \dots, X_M)$ is

$$C_i(X_1, \dots, X_M) := \frac{\partial h(X_1, \dots, X_M)}{\partial X_i} \quad i \in \{1, \dots, M\}$$

Alternatively, we could call this a *structural effect*

Notice that the definition links the term 'causal effect' to the idea of the 'structural model'

Causality therefore is within a particular model

This definition applies for continuous X_i

It's clear how one would tweak this to allow discrete X_i

Can you know C_i ?

Three problems:

- you don't know h
- you don't know all the variables $X_1, \dots, X_M$ that should enter the rhs,
(the known unknowns, and the unknown unknowns)
and even if you did
- you only observe iid copies of $(X_1, \dots, X_K)'$ for $K < M$

How can we help ourselves?

You observe iid copies of Y and $X := (X_1, \dots, X_K)'$

You don't observe $e := (X_{K+1}, \dots, X_M)'$

From your point of view, the structural model becomes $Y = h(X, e)$

(Aside: in the current subsection, the symbol X defines a $K \times 1$ -vector, it will revert back to an $N \times K$ -matrix subsequently)

Knowing about the best predictor property of μ , you consider

$$\begin{aligned}\mu(X) &= \mathbb{E}(h(X_1, \dots, X_M) | X_1, \dots, X_K) \\ &= \mathbb{E}(h(X, e) | X) \\ &= \int h(X, e) \cdot f(e|X) de\end{aligned}$$

and its partial derivative, for $i = 1, \dots, K$,

$$\begin{aligned}\partial\mu(X)/\partial X_i &= \frac{\partial \int h(X, e) \cdot f(e|X) de}{\partial X_i} \\ &= \int \left(\frac{\partial h(X, e)}{\partial X_i} f(e|X) + h(X, e) \frac{\partial f(e|X)}{\partial X_i} \right) de \\ &= \int \left(C_i(X, e) \cdot f(e|X) + h(X, e) \frac{\partial f(e|X)}{\partial X_i} \right) de\end{aligned}$$

The first term gets its own name...

Definition (Average Causal Effect)

The **average causal effect** of X_i on the outcome variable Y in the structural model $Y = h(X_1, \dots, X_M)$ is

$$ACE_i(X) := \int C_i(X, e) \cdot f(e|X) de.$$

It follows that

$$\partial \mu(X) / \partial X_i = ACE_i(X) + \int \left(h(X, e) \frac{\partial f(e|X)}{\partial X_i} \right) de$$

What do we do with the second term on the rhs?

Copy and paste from last slide:

$$\partial\mu(X)/\partial X_i = ACE_i(X) + \int \left(h(X, e) \frac{\partial f(e|X)}{\partial X_i} \right) de,$$

where $i = 1, \dots, K$

Recall that $f(e|X) = f(e|X_1, \dots, X_K)$

What would happen if the distribution of e conditional on $(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_K)$ was independent of X_i ?

Then $f(e|X) = f(e|X_1, \dots, X_K) = f(e|X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_K)$

It follows that under *conditional independence* $\partial f(e|X)/\partial X_i = 0$ and therefore $\partial\mu(X)/\partial X_i = ACE_i(X)$

In words: small deviations of the conditional expectation function identify average causal effects

(This is an important result, but please be aware that we had to pay a high price: we imposed conditional independence)

Exploring some middle ground:

what if we knew that the structural function h was linear?

That is, if $h(X, e) = X'\beta + e$, then

$$\begin{aligned}\mu(X) &= E(h(X, e)|X) \\ &= E(X'\beta + e|X) \\ &= X'\beta + E(e|X)\end{aligned}$$

Studying the partial derivative:

$$\partial\mu(X)/\partial X_i = \partial X'\beta/\partial X_i + \partial E(e|X)/\partial X_i = \beta_i + \partial E(e|X)/\partial X_i$$

What do we do with the second term on the rhs?

Copy and paste from previous slide:

$$\partial\mu(X)/\partial X_i = \partial X'\beta/\partial X_i + \partial E(e|X)/\partial X_i = \beta_i + \partial E(e|X)/\partial X_i$$

Idea: assume **mean independence**: $E(e|X) = E(e) = 0$

This implies $\frac{\partial E(e|X)}{\partial X_i} = 0$

It follows that $\partial\mu(X)/\partial X_i = \beta_i$

So the average causal effect is very simple

Mean independence is a weaker restriction than conditional independence

But we needed to accept the bargain that the structural function is linear

This leads to the linear regression model...

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Definition (Linear Regression Model)

$$Y_i = X_i' \beta + e_i, \quad E(e_i | X_i) = E(e_i) = 0$$

and $EY_i^2 < \infty$ and $E\|X_i\|^2 < \infty$.

Again, the equality $E(e_i | X_i) = E(e_i)$ holds under **mean independence**

Without loss of generality: $E(e_i) = 0$

(the presence of the constant term ensures this, see PS 4)

It implies $E(e_i X_i) = 0$ because: $E(e_i X_i) = E(X_i \cdot E(e_i | X_i)) = 0$

But not the other way round, eg: $e_i = X_i^2$ with $X_i \sim \mathcal{N}(0, 1)$

(then $E(e_i X_i) = E(X_i^3) = 0$ but clearly $E(e_i | X_i) = E(X_i^2 | X_i) = X_i^2 \neq 0$)

In the linear regression model, because $E(e_i X_i) = 0$, we have

$$\beta = \beta^* = E(X_i X_i')^{-1} E(X_i Y_i)$$

(provided $E(X_i X_i')$ is invertible)

In a way, the linear regression model merely says that the projection coefficient is also the structural coefficient of interest

Luckily we know a good estimator for β^* : the OLS estimator

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We've seen that $E(e_i|X_i) = 0$ is stronger (or more restrictive) than $E(e_iX_i) = 0$

So it shouldn't be too surprising that the mean independence condition enables us to do new things

We can study the *finite sample properties* of the OLS estimator:

- $E(\hat{\beta}^{\text{OLS}})$
- $\text{Var}(\hat{\beta}^{\text{OLS}})$

Last week we studied *approximate* mean and variance via the asymptotic distribution

Today we are able to derive the *exact* mean and variance (without the need to resort to the CLT)

The mean independence condition makes this possible

Expected value

Rewrite

$$\begin{aligned}\hat{\beta}^{\text{OLS}} &= (X'X)^{-1}X'Y \\ &= (X'X)^{-1}X'(X\beta + e) \\ &= \beta + (X'X)^{-1}X'e\end{aligned}$$

(implicit definitions of vectors/matrices are the obvious ones)

So that $\hat{\beta}^{\text{OLS}} - \beta = (X'X)^{-1}X'e$

For some reason, people like when $E\hat{\beta}^{\text{OLS}} = \beta$

If that were true then $\hat{\beta}^{\text{OLS}}$ is **unbiased**

We could make this happen by claiming $E((X'X)^{-1}X'e) = 0$

But can we claim this?

Yes we can!

This follows readily using the law of iterated expectation:

$$\begin{aligned} E((X'X)^{-1}X'e) &= E(E((X'X)^{-1}X'e|X)) \\ &= E((X'X)^{-1}X'E(e|X)) \\ &= 0 \end{aligned}$$

The mean independence condition makes the OLS estimator unbiased

Conditional variance

Under mean independence, $E(\hat{\beta}^{\text{OLS}}) = \beta$, and we get

$$\begin{aligned}\text{Var}(\hat{\beta}^{\text{OLS}}|X) &= E((\hat{\beta}^{\text{OLS}} - \beta)(\hat{\beta}^{\text{OLS}} - \beta)'|X) \\ &= E((X'X)^{-1}X'ee'X(X'X)^{-1}|X) \\ &= (X'X)^{-1}X' \cdot E(ee'|X) \cdot X(X'X)^{-1}\end{aligned}$$

To make this look simpler, assume homoskedasticity:

$$E(ee'|X) = \sigma_e^2 I_N$$

It means that the error variances are not functions of X

It follows that $\text{Var}(\hat{\beta}^{\text{OLS}}|X) = \sigma_e^2(X'X)^{-1}$

Unconditional variance

Lemma (Variance Decomposition)

For $P, Q \in L_2$, $\text{Var}(P) = E(\text{Var}(P|Q)) + \text{Var}(E(P|Q))$

In words:

Total variance equals average within-group variance plus variance between-group means (see PS 4)

Therefore, $\text{Var}(\hat{\beta}^{\text{OLS}}) = E(\text{Var}(\hat{\beta}^{\text{OLS}}|X)) + \text{Var}(E(\hat{\beta}^{\text{OLS}}|X))$

But the second term is zero (why?), thus

$$\begin{aligned}\text{Var}(\hat{\beta}^{\text{OLS}}) &= E(\sigma_e^2(X'X)^{-1}) \\ &= \sigma_e^2 \cdot E((X'X)^{-1})\end{aligned}$$

This is the unconditional variance under mean independence and homoskedasticity

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Gauss Markov Theorem

Theorem (Gauss Markov Theorem)

In the linear regression model with homoskedastic errors, amongst all linear estimators that are conditionally unbiased, $\hat{\beta}^{OLS}$ has the lowest variance.

Some people say that OLS is BLUE:

best linear unbiased estimator

That's not very precise, what does 'best' refer to?

Answer: minimal variance

Sketch of proof

- let $\tilde{\beta} := C'Y$ be any other linear unbiased estimator (where C is a $N \times K$ -dimensional matrix based on X)
- similar to above, its conditional variance is $\text{Var}(\tilde{\beta}|X) = \sigma_e^2 C' C$
- define $D' := C' - (X'X)^{-1}X'$
- unbiasedness implies $C'X = I_K$
because: $E(C'Y|X) = E(C'(X\beta + e)|X) = C'X\beta = \beta$
- therefore $D'X = (C' - (X'X)^{-1}X')X = 0$
- then $C'C = (D' + (X'X)^{-1}X')(D' + (X'X)^{-1}X')' = D'D + (X'X)^{-1}$
- it follows

$$\begin{aligned}\text{Var}(\tilde{\beta}|X) &= \sigma_e^2 C' C = \sigma_e^2 ((X'X)^{-1} + D'D) \\ &\geq \sigma_e^2 (X'X)^{-1} \\ &= \text{Var}(\hat{\beta}^{\text{OLS}}|X),\end{aligned}$$

inequality because $D'D$ is nonnegative definite (next slide)

Definition (Nonnegative definiteness)

A symmetric matrix P is nonnegative definite if $q'Pq \geq 0$ for all vectors q .

Lemma

$D'D$ is nonnegative definite.

Sketch of proof

D is an $N \times K$ matrix, so $D'D$ is $K \times K$

Take $K \times 1$ vector q , then $q'D'Dq = (q'D')(Dq) = (Dq)'(Dq) \geq 0$

The rhs has the form $r'r$ where r is $N \times 1$

$r'r$ is obviously nonnegative

A constructive proof of Gauss Markov Theorem
(from Amemiya's textbook)

Let $\tilde{\beta} := C'Y$ be any other linear unbiased estimator,
where C is a $N \times K$ -dimensional matrix based on X

Under homoskedasticity, $\text{Var}(\tilde{\beta}|X) = \sigma_e^2 C' C$

Unbiasedness implies $C'X = I$

because: $E(C'Y|X) = E(C'(X\beta + e)|X) = C'X\beta = \beta$

Let's come up with some way of minimizing the variance given the
'constraint' that the estimator is unbiased

We set up a scalar minimization problem

For that, let p be an arbitrary K -vector

(its purpose is to pick any particular linear combination of $\tilde{\beta}$)

Instead of considering the vector $\tilde{\beta}$ we look at the scalar $p'\tilde{\beta}$

If $\tilde{\beta} = C'Y$ is an estimator of β ,

then $p'C'Y$ is the corresponding estimator of $p'\beta$

Define $c := Cp$, so that $p'C'Y = c'Y$

and notice that unbiasedness implies $X'c = X' Cp = p$

Also, $\text{Var}(p'\tilde{\beta}|X) = \text{Var}(c'Y|X) = \sigma_e^2 c'c$

Our minimization problem is:

Minimize $\text{Var}(c'Y|X)$ subject to the unbiasedness constraint $X'c = p$

The corresponding Lagrangian is $L(c, \lambda) = c'c - 2\lambda'(X'c - p)$

Next: take derivative with respect to c and solve

Remember the following useful matrix derivative rules:

$$\frac{\partial Az}{\partial z} = A'$$

$$\frac{\partial(z'Az)}{\partial z} = (A + A')z$$

Therefore,

$$\frac{\partial(c'c - 2\lambda'(X'c - p))}{\partial c} = 2c - 2X\lambda$$

Setting equal to zero results in $c = X\lambda$ or, alternatively, $X'c = X'X\lambda$

Using the constraint $X'c = p$ gives $X'X\lambda = p$ or, alternatively,
 $\lambda = (X'X)^{-1}p$

And finally, $c = X\lambda = X(X'X)^{-1}p$, so that the minimum variance unbiased estimator of $p'\beta$ turns out to be $p'(X'X)^{-1}X'Y$

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In the last assignment you have learned that the *generalized least squares estimator* $\hat{\beta}^{\text{GLS}}$ is the minimum variance unbiased estimator in the linear regression model under heteroskedasticity

This is a Gauss Markov theorem for the heteroskedastic case

But the derivation assumed knowledge of $E(ee'|X)$

In real life you don't have that knowledge, and $\hat{\beta}^{\text{GLS}}$ is practically useless, you cannot calculate it

For that reason we call $\hat{\beta}^{\text{GLS}}$ the *infeasible* GLS estimator

There exists a *feasible* variant, but it isn't used much

Let's first revisit the GLS setup

Error variance was

$$\begin{aligned} E(ee'|X) &= \sigma^2 \cdot \Gamma = \sigma^2 \cdot \text{diag}(\gamma_1, \dots, \gamma_N) \\ &= \text{diag}(\sigma_1^2, \dots, \sigma_N^2) =: \Sigma \end{aligned}$$

Define $\tilde{Y} := \Gamma^{-1/2}Y$ and $\tilde{X} := \Gamma^{-1/2}X$

The GLS estimator is motivated as the OLS estimator of $\tilde{Y}$ on $\tilde{X}$:

$$\begin{aligned} \hat{\beta}^{\text{GLS}} &:= (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y} \\ &= (X'\Gamma^{-1}X)^{-1}X'\Gamma^{-1}Y \\ &= (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}Y \\ &= \left(\sum_{i=1}^N X_i X_i' / \sigma_i^2 \right)^{-1} \sum_{i=1}^N X_i Y_i / \sigma_i^2 \end{aligned}$$

Each observation is weighed inversely to its error variance

Hence the alternative name *weighted least squares* estimator

How could we turn *infeasible* GLS into a *feasible* estimator?

Idea: use $\hat{\Sigma} = \Sigma + o_p(1)$ in place of Σ

Where does this consistent variance estimator come from?

Easy: OLS will provide a consistent (yet inefficient) estimator of β and therefore also of Σ

This suggests the following two step estimation approach:

- (i) run OLS of Y on X , compute $\hat{e}$ and obtain $\hat{\Sigma}$ by imposing some structure on $E(e_i^2|X_i)$, for example $E(e_i^2|X_i) = \sigma(X_i)$ where σ is some known function
- (ii) compute $\hat{\beta}_{\text{feas}}^{\text{GLS}} := (X'\hat{\Sigma}^{-1}X)^{-1}X'\hat{\Sigma}^{-1}Y$

Nobody uses this in practice, it's a textbook-only estimator

Feasible GLS does not satisfy the Gauss Markov theorem
(because using $\hat{\Sigma}$ instead of Σ adds sampling error)